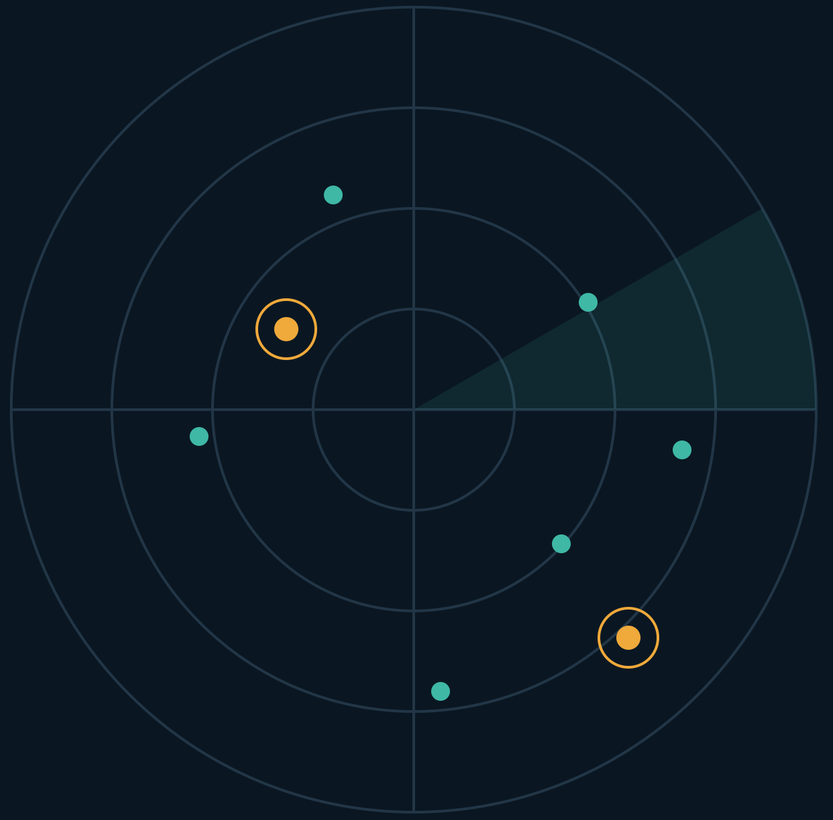
SUPERNOVA × SPACEPOINT · THEME 03 · SEAS OF DATA

Radar sees every ship. We find the ones that hide.



WORKING PROTOTYPE

REAL SENTINEL-1 DATA

LIVE DASHBOARD

BUOY CAD MODEL

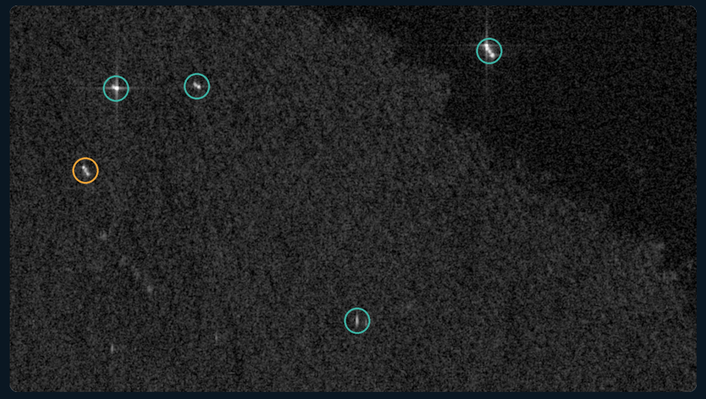
Team Interstellar · Tohraan Ahmad Khan · Ayush Khale · Aryan Khale · Mayank

BITS Pilani, Dubai Campus · November 2026

It already runs: real radar in, flagged ships and a buoy model out

Detector

RUNNING

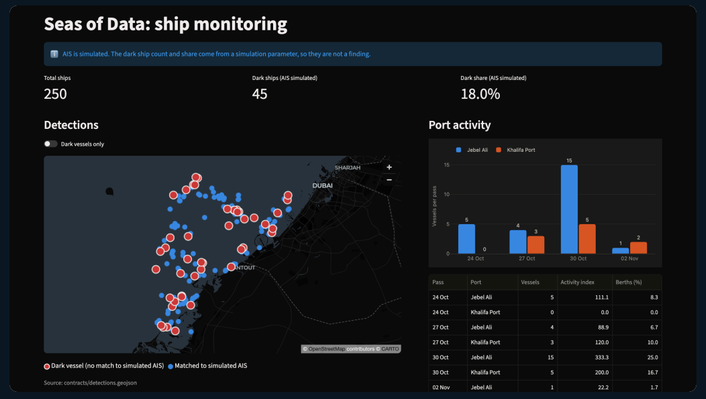


4
REAL PASSES

250
SHIPS FOUND

Dashboard

RUNNING

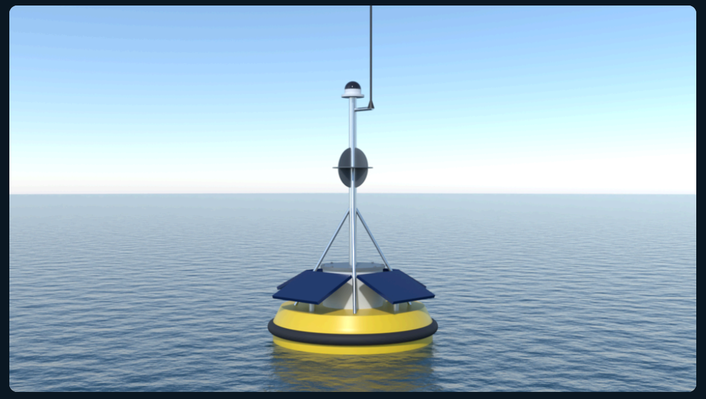


250
ON THE MAP

2
PORTS COUNTED

Buoy

CAD BUILT



14
CAD PARTS

4.5 m
TALL

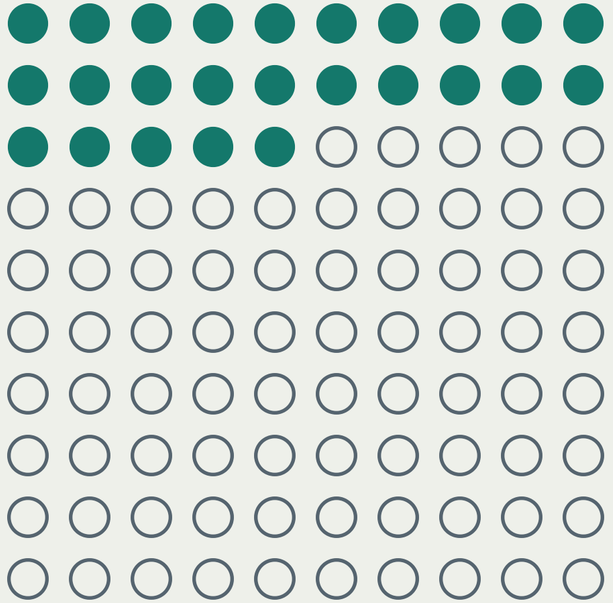
● REAL SATELLITE DATA ● REAL DETECTOR ○ AIS SIMULATED ○ BUOY NOT YET BUILT

About three in four industrial fishing vessels are missing from public ship tracking

72–76%

of the world’s industrial fishing vessels are not publicly tracked.

Global figure, 2017 to 2021, not UAE-specific. Dark does not mean illegal.



Each circle = 1 of 100 vessels.
Filled = tracked.

SOURCE: PAOLO ET AL., NATURE 625, 85-91 (2024)

Sentinel-1 passes over Jebel Ali about every three days

COVERAGE

250 km swath, 10 m pixels, VV band

CONDITIONS

Day, night and through cloud

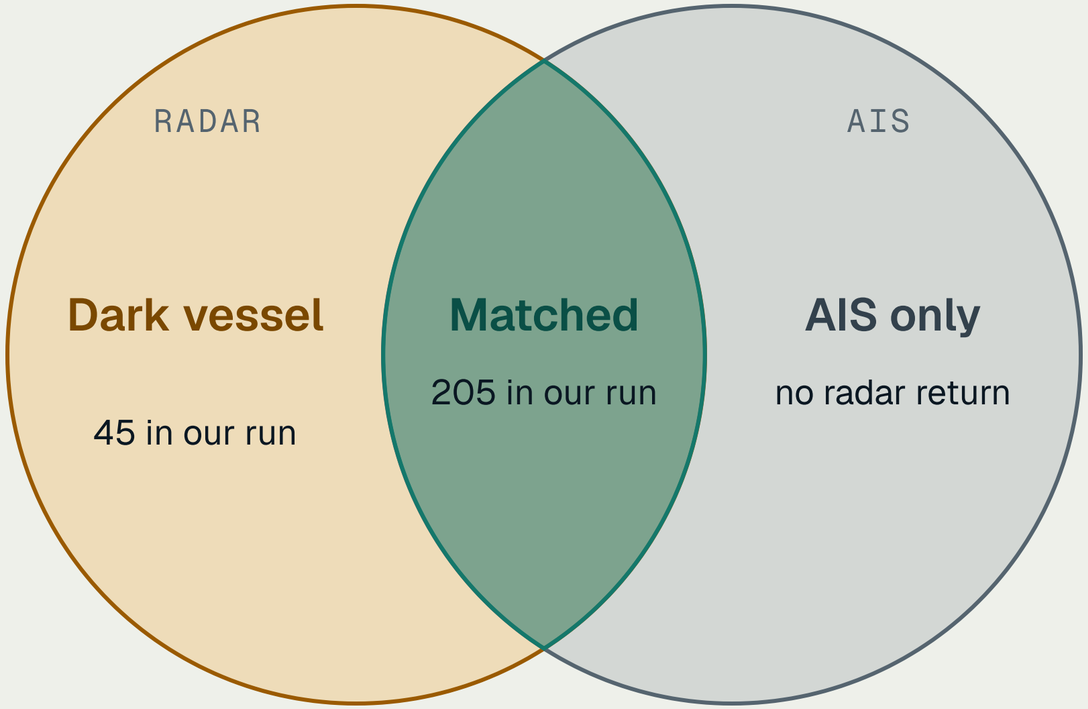
ACCESS

Free and open through Copernicus



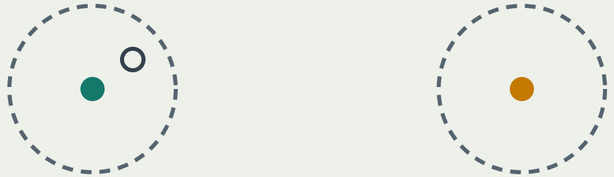
PASSES FOUND IN THE COPERNICUS CATALOGUE · 25.01°N 55.06°E

Radar minus AIS equals the dark vessels



MATCH GATE

500 m



MATCHED DARK

Dark is a lead for patrols, not a verdict.

AIS SIMULATED IN THIS PROTOTYPE

One system: the satellite finds, the dashboard shows, the buoy confirms

01 · SPACE — REAL DATA

Sentinel-1 radar

Free, every three days, any weather

02 · PIPELINE — RUNNING

Detect and match

Sea mask, CFAR, coordinates, AIS match

03 · DASHBOARD — RUNNING

See and act

Map, dark flags, port counts

DARK-SHIP ALERT

AIS + PHOTO

04 · SEA — CAD BUILT

Solar buoy

AIS log, camera, oil sensor

THE LOOP — DESIGNED

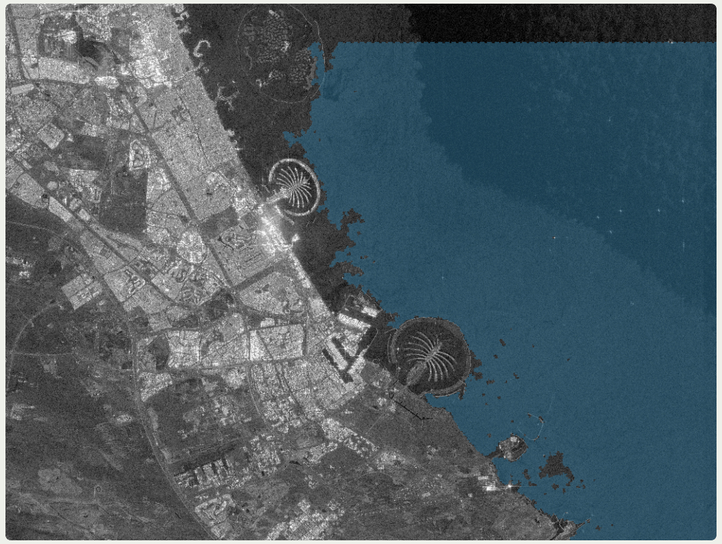
1. Radar flags a ship with no AIS report
2. The nearest buoy checks it with AIS and camera
3. Real AIS replaces the simulated list

Real radar in, ships out, in three steps



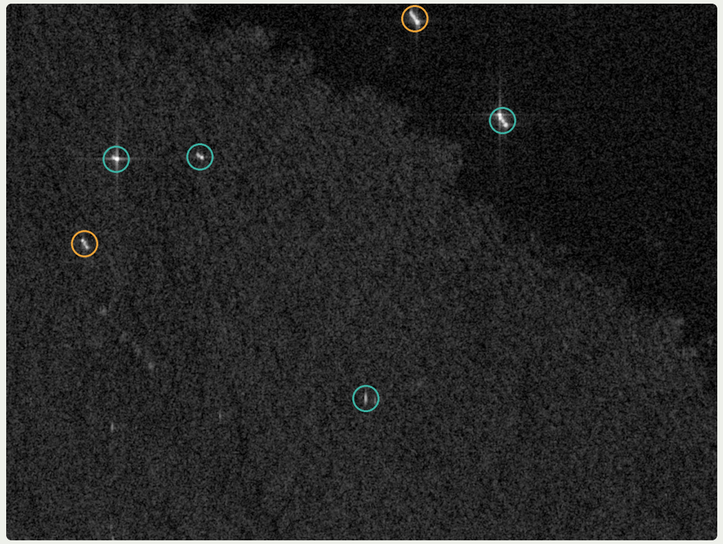
1 · RAW SCENE

2 Nov 2026. Bright = land and hulls.



2 · SEA MASK

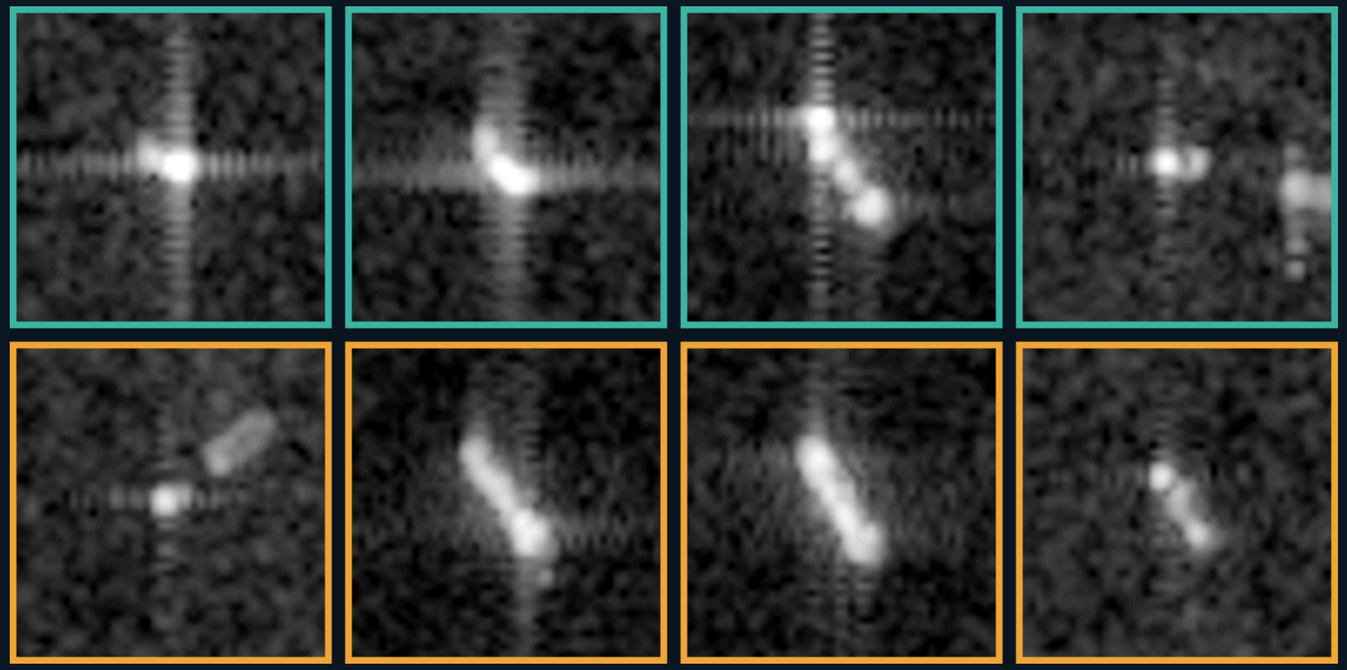
Blue = sea. Coast mapped once, reused.



3 · DETECTIONS

Teal ring = AIS match. Amber = none.

Close up: teal ships have an AIS report, amber ones do not



- 480 m square around each ship
- Top row: AIS match
- Bottom row: no AIS match, so dark

Four real passes: 250 ships located, 45 with no simulated AIS match



2 NOV 2026 · 02:14 UTC · 20 OF 73 SHIPS SHOWN

250
SHIPS

205
MATCHED

45
DARK

AIS IS SIMULATED

The dark share is our simulation, not a finding about Gulf shipping.

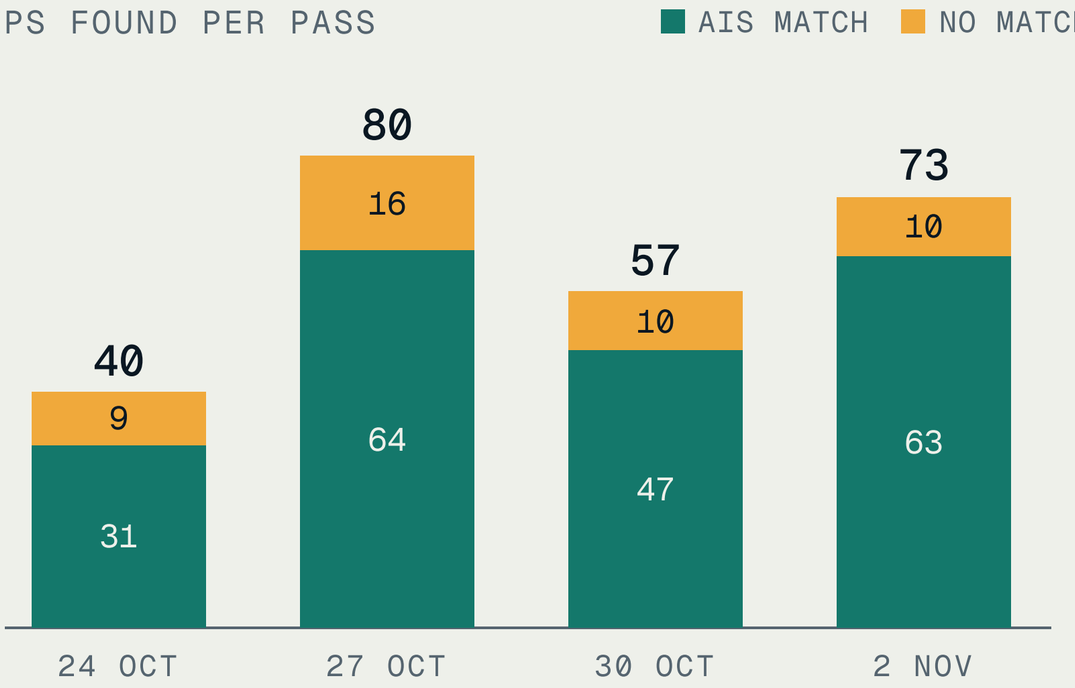
HOW TO READ IT

Teal ring: AIS match. Amber ring: none, a lead to check.

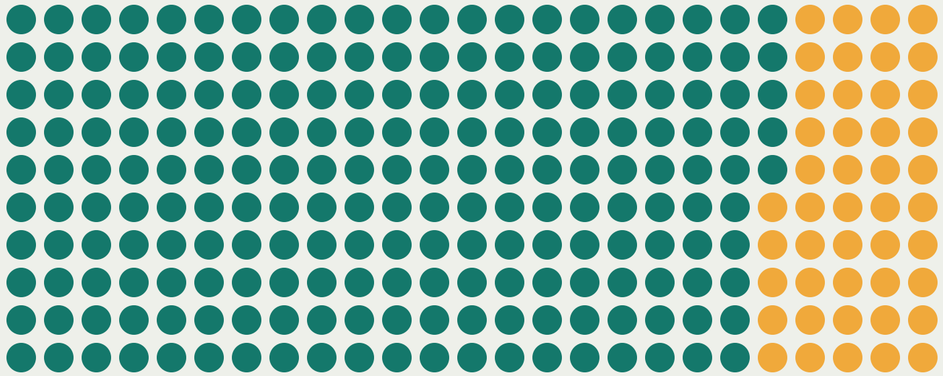
SCENE · SEA MASK · CFAR · MERGE · GEOCODE · AIS MATCH 500 M

Each pass finds 40 to 80 ships; about one in five has no simulated AIS match

SHIPS FOUND PER PASS



ALL 250 SHIPS, ONE DOT EACH



205 matched **45 no match**

AIS is simulated: the 45 shows the method, not Gulf shipping.

One screen: every ship, dark ones flagged, ships counted per port



RUNNING

1 **Map**

Red: no AIS match. Blue: matched.

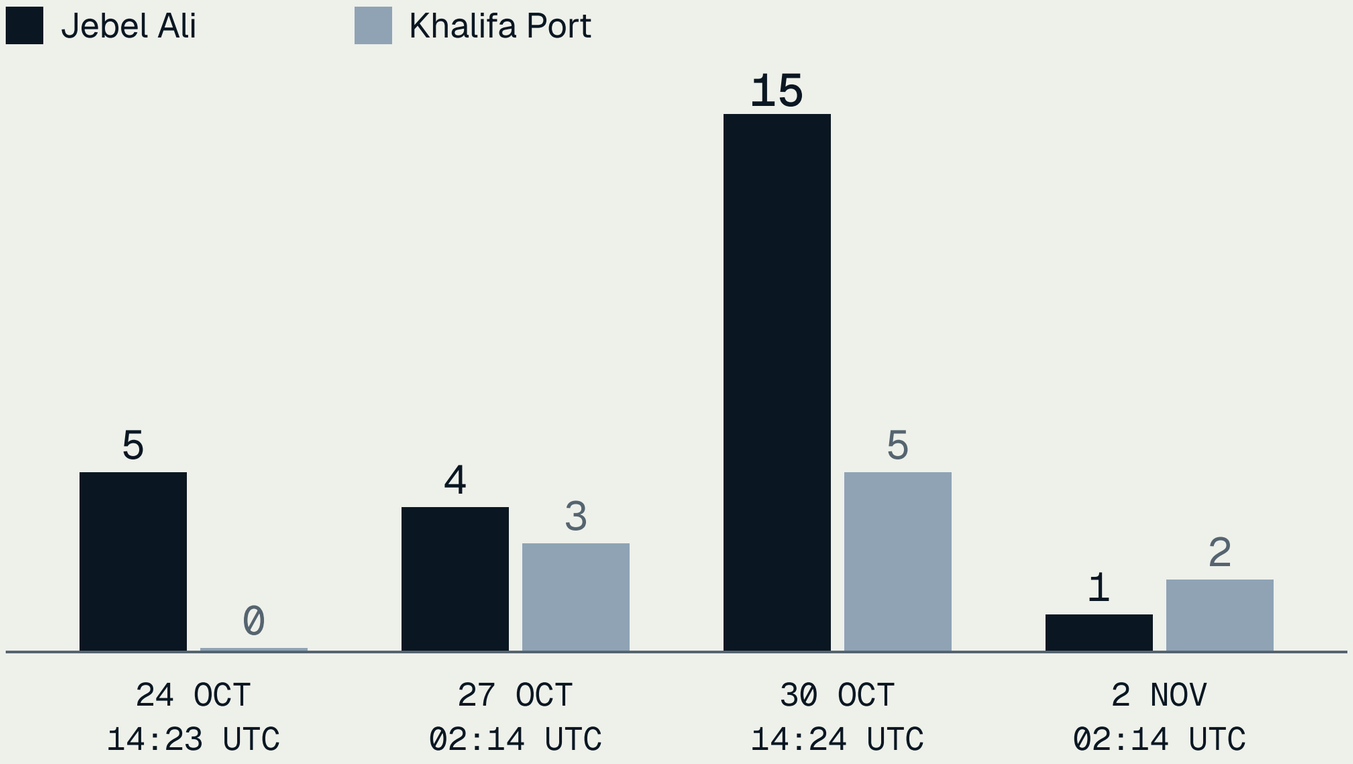
2 **Banner**

AIS is simulated, said everywhere.

3 **Port panel**

Ship counts per port and pass.

Four passes give a baseline, not a trend: Jebel Ali’s approaches held 1 to 15 ships



THE INDEX

Count ÷ port median × 100

WHAT THE COUNTS ARE

Ships at anchor and in the approaches

WHY NO TREND YET

Sea state shifts what radar sees; not corrected yet

SHIPS DETECTED PER PORT BOX · 4 PASSES

On 1,962 radar images it never saw, the detector finds 82% of ships at 91% precision

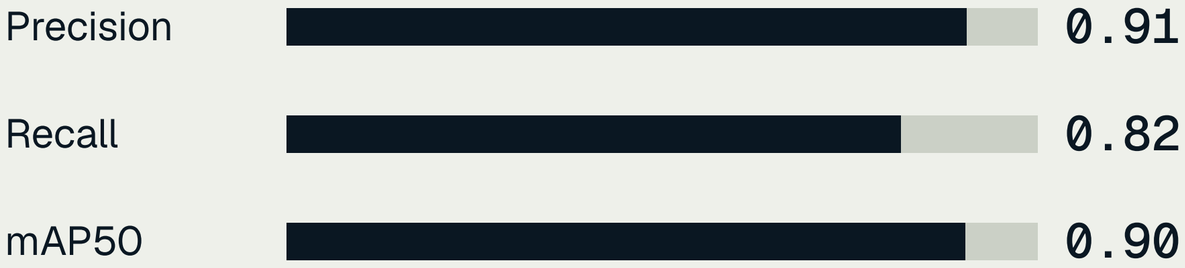
5,604 HRSID IMAGES, SPLIT THREE WAYS



3,278 train

364 tune

1,962 never seen



Scores on the held-out test split. Bars run from 0 to 1.

WHY IT MATTERS

Misses hide ships. False alarms invent them.

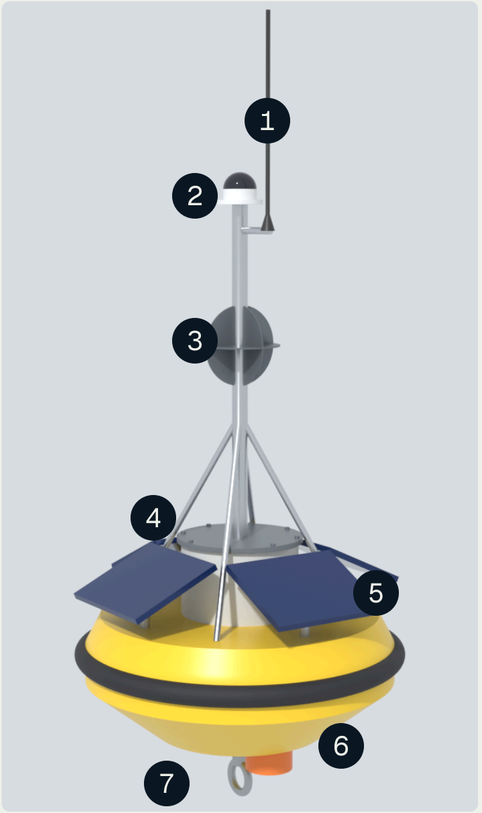
WHAT THIS IS NOT

Benchmark pixels are finer than 10 m Sentinel-1. Gulf accuracy is not measured.

WHAT WE DID ABOUT IT

Gulf passes use a CFAR detector that needs no training.

A solar buoy checks what radar flags, between satellite passes



1 **AIS antenna**
Hears ships that broadcast

2 **Camera dome**
Sees silent ships

3 **Radar reflector**
Shows up on ships' radar

4 **Electronics bay**
Battery, processor, radio

5 **Four solar panels**
Power, tilted to shed spray

6 **Oil-sheen sensor**
Flags possible oil

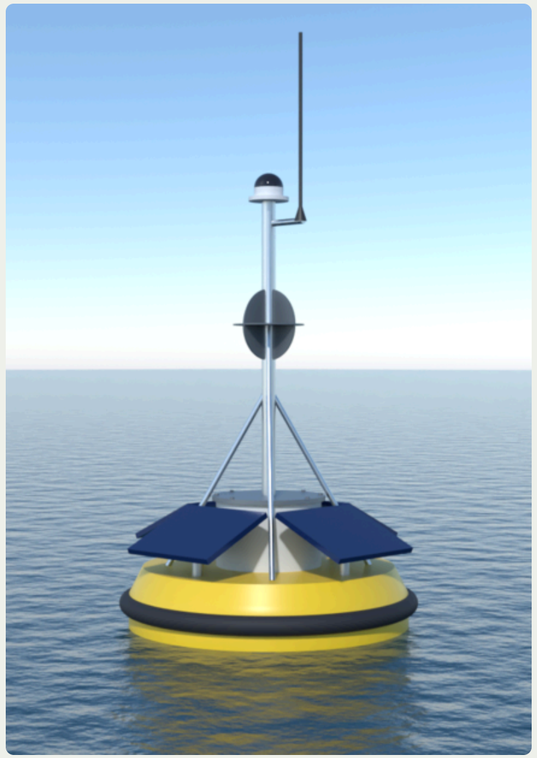
7 **Mooring eye**
Anchor chain point

CAD BUILT

4.5 m TALL	1.8 m HULL Ø
14 CAD PARTS	4 SOLAR PANELS

Design only: not built, not tested afloat.

One CAD model, four Blender views



3D VIEW

Hull, panels and mast together



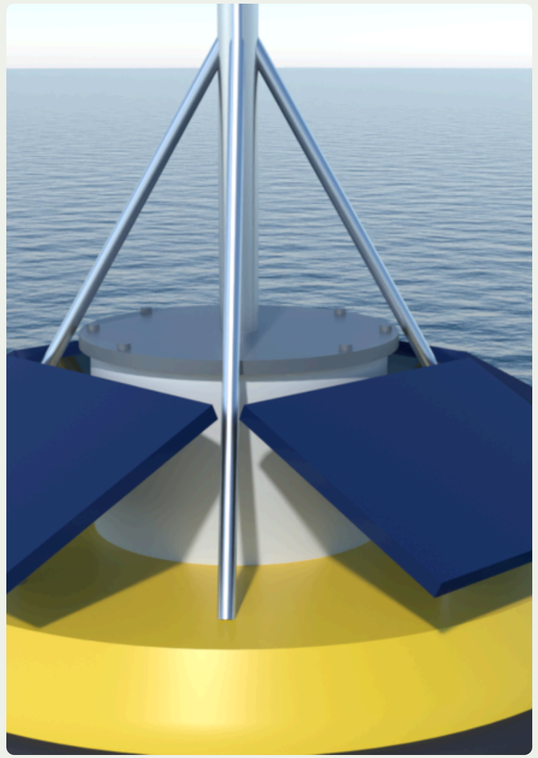
SIDE

Panels tilted 25°, sensor below



FROM ABOVE

Four panels, one per side

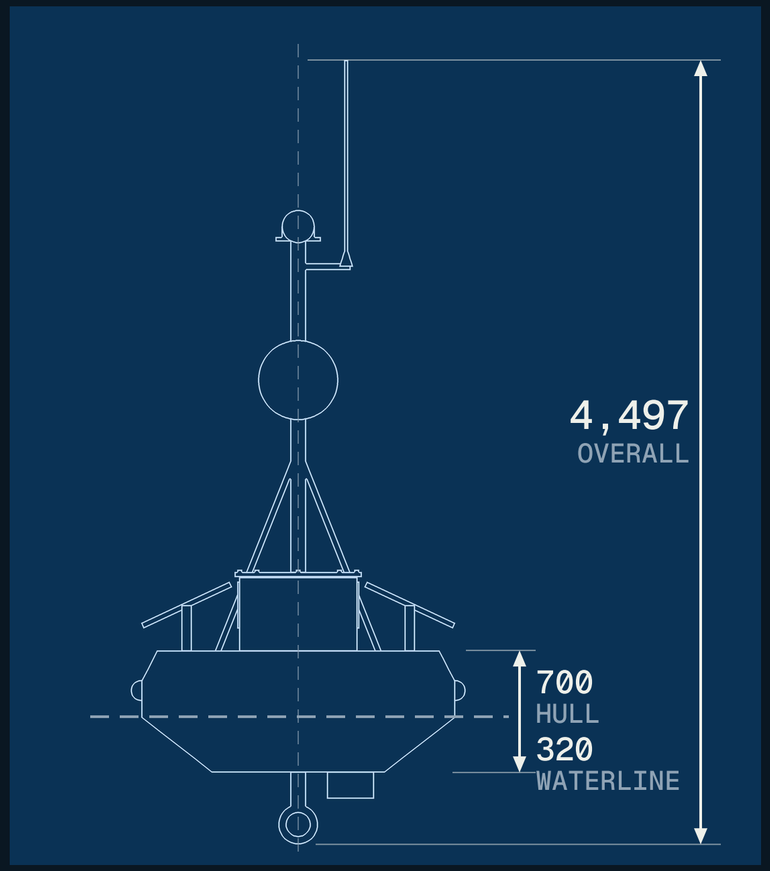


MAST HEAD

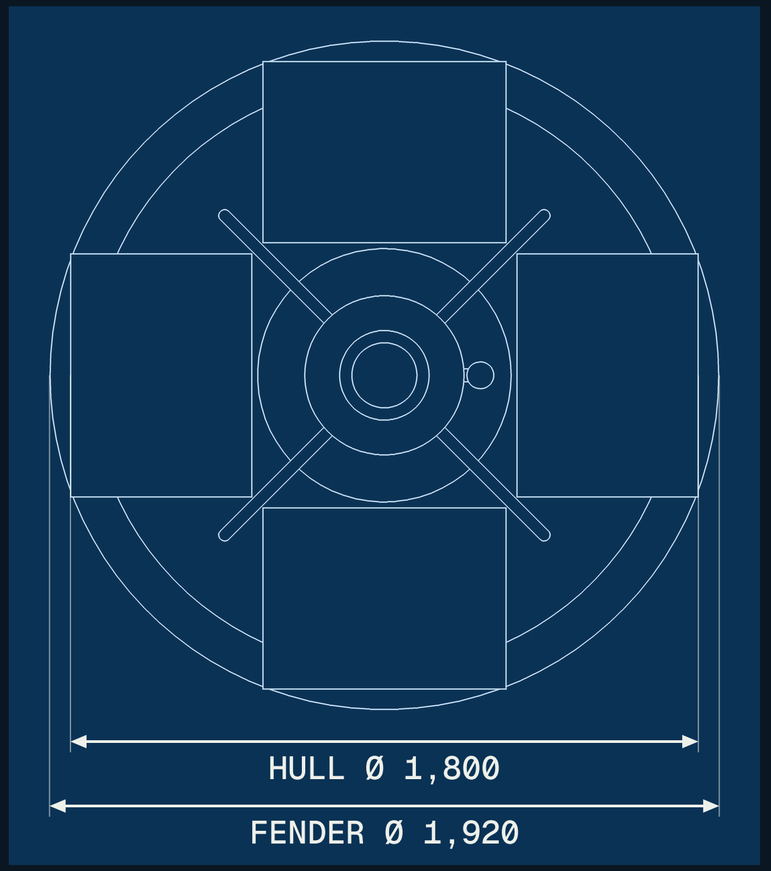
Camera, AIS and reflector

On paper: the buoy is 4.5 m tall and 1.8 m across

SIDE ELEVATION



PLAN VIEW



OVERALL HEIGHT	4,497 mm
HULL Ø	1,800 mm
HULL HEIGHT	700 mm
WATERLINE	320 mm
PANEL TILT	25°

DRAWN FROM THE CAD MESHES.
WATERLINE IS A DESIGN TARGET,
NOT A CALCULATION.

What is real, simulated, and still to come

REAL AND RUNNING

- 4 Sentinel-1 passes
- Ship detector, sea mask
- Live dashboard
- Buoy CAD model

SIMULATED OR ASSUMED

- AIS reports
- Port capacity
- Buoy afloat: not built or tested

NEXT

- Connect a real AIS feed
- Retrain on xView3-SAR
- Add oil-spill detection

Data, models and licences

Component	Used for	Source and terms
Sentinel-1 GRD	Real Gulf scenes, VV band	ESA, via Copernicus Data Space, free and open
HRSID	Training and held-out test of the ship detector	Wei et al., IEEE Access; repository under GPL-3.0
YOLO11	Detector, fine-tuned from pretrained weights	Ultralytics
Port boxes	Counting ships inside Jebel Ali and Khalifa Port	Drawn by us, checked against OpenStreetMap extents

Contains modified Copernicus Sentinel data 2026.

Method and parameters

Step	Setting	Reason
Sea mask	Static lon/lat grid, 0.001°, built from a calm-sea pass	Windy sea is as bright as desert, so brightness cannot find the coast
Detector	CFAR, 61 px window, 8 sigma, at least 3 px, at most 450 m	Needs no training and works at 10 m pixels
Merge	Returns within 150 m become one ship	One hull often splits into several blobs
Geocoding	Ground control points in each GRD file	Checked against Palm Jumeirah and Palm Jebel Ali
AIS match	Simulated: 80% reported, 120 m jitter; gate 500 m	No free real Gulf AIS feed
YOLO11s	Fine-tuned on HRSID, scored on 1,962 held-out images: precision 0.91, recall 0.82, mAP50 0.90	Benchmark score only; Gulf passes use CFAR